

## LEONORA AP, Australia

WMO: 944500

Lat: 28.879S

Lon: 121.319E

Elev: 371

StdP: 96.95

Time Zone: 8.00 (E08)

Period: 07-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 7             | 3.1        | 4.4 | -10.6                         | 1.6 | 28.7 | -9.0 | 1.8 | 26.3 | 10.2                  | 19.0 | 9.3 | 17.9 | 1.0                   | 110  | 0.491 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 12.7                   | 41.8            | 18.3 | 40.3 | 18.1 | 38.6 | 18.0 | 22.7                | 29.7 | 21.8 | 30.1 | 20.9 | 30.0 | 5.1                  | 250  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 20.9                            | 16.3 | 26.0 | 19.6 | 15.0 | 24.9 | 18.2 | 13.7 | 24.3 | 69.4          | 29.7 | 65.5 | 29.9 | 62.2 | 30.2 | 25.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 9.6               | 8.6  | 7.9 | 0.3                        | 44.3 | 0.8                | 1.4 | -0.3                                               | 45.3 | -0.8       | 46.1 | -1.2       | 46.9 | -1.9       | 48.0 |
|                   |      |     | -1.1                       | 24.0 | 1.0                | 1.1 | -1.8                                               | 24.7 | -2.4       | 25.4 | -3.0       | 26.0 | -3.8       | 26.8 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 21.2   | 29.7  | 28.1  | 25.2  | 21.7  | 16.3  | 13.2  | 12.4  | 14.4  | 17.4  | 22.7  | 25.2  | 28.0  |  |  |
|                                                                        | DBStd       | 7.02   | 4.22  | 4.42  | 4.04  | 3.50  | 3.09  | 2.77  | 2.77  | 3.32  | 3.78  | 4.04  | 4.10  | 4.07  |  |  |
|                                                                        | HDD10.0     | 13     | 0     | 0     | 0     | 0     | 0     | 3     | 8     | 2     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | HDD18.3     | 632    | 1     | 1     | 2     | 7     | 79    | 156   | 183   | 129   | 63    | 9     | 2     | 0     |  |  |
|                                                                        | CDD10.0     | 4085   | 611   | 507   | 471   | 351   | 195   | 99    | 83    | 139   | 222   | 392   | 456   | 558   |  |  |
|                                                                        | CDD18.3     | 1663   | 353   | 274   | 214   | 109   | 16    | 2     | 1     | 7     | 36    | 143   | 208   | 300   |  |  |
|                                                                        | CDH23.3     | 21841  | 5165  | 3715  | 2497  | 1073  | 169   | 11    | 15    | 112   | 507   | 1764  | 2647  | 4166  |  |  |
| (13)                                                                   | CDH26.7     | 12214  | 3277  | 2237  | 1278  | 427   | 34    | 0     | 1     | 17    | 179   | 851   | 1429  | 2483  |  |  |
| (14)                                                                   | Wind        | WSAvg  | 4.1   | 5.1   | 4.8   | 4.5   | 3.6   | 3.3   | 3.2   | 3.7   | 4.0   | 4.5   | 4.8   | 4.9   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 249    | 19    | 42    | 27    | 21    | 27    | 26    | 23    | 16    | 10    | 13    | 14    | 19    |  |  |
|                                                                        | PrecMax     | 552    | 128   | 285   | 179   | 101   | 110   | 100   | 102   | 79    | 47    | 56    | 52    | 94    |  |  |
|                                                                        | PrecMin     | 94     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | PrecStd     | 109    | 28    | 59    | 39    | 23    | 28    | 24    | 26    | 18    | 11    | 16    | 13    | 22    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 44.5  | 43.2  | 39.9  | 36.4  | 29.8  | 24.7  | 25.3  | 29.1  | 33.6  | 37.7  | 41.4  | 42.2  |  |  |
|                                                                        |             | MCWB   | 19.0  | 18.3  | 17.6  | 17.3  | 14.3  | 13.3  | 12.8  | 12.6  | 14.1  | 16.0  | 16.9  | 18.0  |  |  |
|                                                                        | 2%          | DB     | 42.2  | 41.0  | 37.4  | 33.4  | 27.2  | 22.5  | 22.7  | 26.4  | 31.0  | 35.9  | 38.5  | 40.8  |  |  |
|                                                                        |             | MCWB   | 18.7  | 18.2  | 17.9  | 16.7  | 13.5  | 12.3  | 12.1  | 12.3  | 13.7  | 15.6  | 16.6  | 17.6  |  |  |
|                                                                        | 5%          | DB     | 40.5  | 39.1  | 35.4  | 31.2  | 25.1  | 20.8  | 20.7  | 24.2  | 28.8  | 33.7  | 36.2  | 38.9  |  |  |
|                                                                        |             | MCWB   | 18.7  | 18.4  | 17.6  | 16.3  | 13.5  | 11.7  | 11.3  | 11.7  | 13.1  | 15.1  | 16.1  | 17.6  |  |  |
|                                                                        | 10%         | DB     | 38.5  | 36.8  | 33.2  | 29.0  | 23.2  | 19.1  | 19.0  | 22.0  | 26.3  | 31.7  | 33.8  | 36.6  |  |  |
|                                                                        |             | MCWB   | 18.5  | 18.6  | 17.6  | 15.8  | 13.1  | 11.1  | 10.6  | 11.4  | 12.6  | 14.6  | 15.8  | 17.3  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 24.1  | 24.1  | 21.8  | 20.2  | 17.4  | 15.5  | 16.2  | 15.4  | 16.0  | 18.6  | 19.9  | 22.4  |  |  |
|                                                                        |             | MCDB   | 30.5  | 31.2  | 28.1  | 28.1  | 23.0  | 20.2  | 21.2  | 21.5  | 26.3  | 28.1  | 28.4  | 31.3  |  |  |
|                                                                        | 2%          | WB     | 22.9  | 22.9  | 20.7  | 18.7  | 15.9  | 14.4  | 13.8  | 14.1  | 14.8  | 17.3  | 19.0  | 21.1  |  |  |
|                                                                        |             | MCDB   | 29.2  | 29.6  | 28.3  | 27.2  | 22.8  | 19.1  | 18.7  | 20.7  | 26.5  | 28.0  | 28.6  | 30.6  |  |  |
|                                                                        | 5%          | WB     | 21.8  | 22.0  | 20.0  | 17.9  | 14.9  | 13.4  | 12.8  | 13.2  | 14.0  | 16.5  | 18.1  | 19.9  |  |  |
|                                                                        |             | MCDB   | 31.2  | 29.2  | 28.2  | 26.1  | 22.0  | 18.2  | 17.8  | 20.9  | 25.9  | 28.4  | 29.7  | 31.0  |  |  |
|                                                                        | 10%         | WB     | 20.9  | 21.0  | 19.2  | 17.1  | 13.9  | 12.4  | 11.8  | 12.2  | 13.2  | 15.7  | 17.3  | 18.7  |  |  |
|                                                                        |             | MCDB   | 31.4  | 29.3  | 28.0  | 25.2  | 21.2  | 17.0  | 17.2  | 20.2  | 24.7  | 28.9  | 29.6  | 32.2  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 12.7  | 12.0  | 11.3  | 12.1  | 12.5  | 11.2  | 12.2  | 13.3  | 14.0  | 13.7  | 13.2  | 13.0  |  |  |
|                                                                        |             | MCDBR  | 15.0  | 14.7  | 14.2  | 14.5  | 14.7  | 13.4  | 15.0  | 16.1  | 17.7  | 16.4  | 15.9  | 14.7  |  |  |
|                                                                        |             | MCWBR  | 3.9   | 4.0   | 4.2   | 4.4   | 5.9   | 6.2   | 7.1   | 6.9   | 7.0   | 5.7   | 5.1   | 4.5   |  |  |
|                                                                        | 5% WB       | MCDBR  | 12.2  | 11.3  | 10.8  | 11.0  | 11.5  | 10.0  | 11.0  | 12.4  | 14.4  | 13.7  | 13.4  | 13.3  |  |  |
|                                                                        |             | MCWBR  | 4.0   | 3.9   | 3.8   | 4.3   | 4.9   | 5.3   | 6.1   | 6.0   | 6.2   | 5.2   | 4.4   | 4.5   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.342 | 0.345 | 0.325 | 0.312 | 0.289 | 0.278 | 0.276 | 0.283 | 0.301 | 0.321 | 0.328 | 0.331 |  |  |
|                                                                        | taud        |        | 2.579 | 2.590 | 2.621 | 2.629 | 2.653 | 2.665 | 2.660 | 2.618 | 2.556 | 2.513 | 2.529 | 2.567 |  |  |
|                                                                        | Ebn at Noon |        | 999   | 979   | 969   | 935   | 914   | 904   | 920   | 952   | 978   | 991   | 1006  | 1013  |  |  |
|                                                                        | Edh at Noon |        | 106   | 102   | 94    | 84    | 74    | 69    | 72    | 83    | 98    | 109   | 111   | 108   |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 7.80  | 6.93  | 6.01  | 4.87  | 3.92  | 3.31  | 3.65  | 4.71  | 6.08  | 7.07  | 7.83  | 8.19  |  |  |
|                                                                        | RadStd      |        | 0.62  | 0.57  | 0.43  | 0.26  | 0.25  | 0.21  | 0.24  | 0.26  | 0.24  | 0.44  | 0.34  | 0.46  |  |  |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only           |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (3 neighbors) |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page